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AI-Powered Companion for Asha Worker

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ABSTRACT:

The problem of poor internet connectivity and digital infrastructure makes it difficult to maintain patient records in several rural and remote regions. At the community level, Accredited Social Health Activists (ASHA workers) are vital in offering basic healthcare services. They usually use paper-based records that lead to reporting problems such as delay, loss of data, and duplication. For ASHA staff in low-network areas, this research recommends the development of an electronic health record (EHR) companion system on mobile devices. This system will allow ASHA staff to enter, retain, and manipulate patient health information in a mobile application in online and offline modes. In low network connectivity areas, the patient records will be kept locally by the system. When the connectivity is back, the records will be synced to a central healthcare database. This means that patients' data will be collected seamlessly. Some of the functions included in this system are patient registration, maternal and child healthcare information, vaccination, and health monitoring. The proposed system seeks to reduce the workload of ASHA workers by enhancing efficiency and accuracy of healthcare data management through reliable digital record-keeping in offline mode.

Keywords: *Electronic Health Records, ASHA Workers, Mobile Health (mHealth), Rural Healthcare, Offline Data Synchronization, Low Network Connectivity.*

1. Introduction

Healthcare delivery in rural India confronts a profound paradox: While the nation boasts the world's largest network of community health workers, with over 1 million ASHA workers bridging the gap between the community and healthcare system, the digital technologies used by those workers are repeatedly unsuccessful not for a lack of ability among the workers, but because they simply aren't designed properly for the realities of the job. ASHA workers make home visits through far-flung, dispersed villages, sometimes traversing distances of several kilometres from household to household. They enter patient data – haemoglobin level, blood pressure, pregnancy status, symptoms, immunization history – onto paper records, which suffer from issues of loss, deterioration, and even illegibility. The introduction of digital devices tends to make the same mistakes: assuming the presence of stable internet connectivity, that English will be used, and that the workers already have basic technical skills.

Care Connect was designed to solve this problem. It is a web-based, mobile-first platform which works both with and without internet connectivity, takes automatically transcribed voice input in regional Indian languages, automatically identifies at-risk patients, provides automatic vaccinations and ANC check-up reminders, and offers individual dashboards for ASHA workers and PHCs alike. This paper details the design, technical structure, and features implemented in Care Connect as a replicable blueprint.

2. Problem Statement

Three main areas of problem can be highlighted:

2.1 Connectivity Dependency

Connectivity Dependency In most cases, the digital health record systems depend on a constant internet connection for proper functioning. This means that in rural India, due to the absence of broadband infrastructure, except in tier-2 and tier-3 urban towns, the use of mobile data connectivity can be sporadic. In the case of ASHAs, when visiting a home in a remote village, it is impossible to use a cloud system for the collection and storage of patient health data.

2.2 Language and Literacy Challenges

It is estimated that the majority of ASHAs speak in languages such as Hindi, Marathi, Tamil, Telugu, Bengali, and Kannada, among others. In digital interfaces that prioritize English, language barriers reduce user engagement and increase data input errors. Inputting text using the keyboard poses an additional challenge, especially since ASHAs might lack experience with mobile devices' typing functionality.

2.3 Absence of Decision Support

ASHAs are community health volunteers trained to assist with the provision of basic healthcare services. Decision support systems would help ASHAs in understanding the need for emergency intervention when certain health conditions are observed. For example, ASHAs should be able to recognize that a pregnant woman might have critically low haemoglobin levels and high blood pressure

3. Literature Review

3.1 mHealth in Low-Resource Settings

According to WHO (2016), mobile health, or mHealth, involves using mobile and wireless technologies to support the attainment of health goals. Since the formal definition of mHealth in the year 2006, the approach has gained momentum in LMICs, having developed into a key strategy for promoting public health in low-resource environments. mHealth applications during their early days were mainly SMS based. Research carried out in Kenya, India, Bangladesh, and Uganda showed significant improvement in prenatal care attendances, child immunization coverage, and adherence of HIV-positive individuals to prescribed treatment following the application of SMS-based reminders (Lester et al., 2010; Labrecque et al., 2013). However, there was a major caveat regarding such applications; they would work only when the application itself was adapted to fit the user's context. Technologies that needed a smartphone, internet connectivity, or literacy in English had a problem reaching their targeted populations. As smartphones became more widely available and mobile data networks expanded even into remote locations, the complexity of mHealth applications began increasing. Apps moved beyond reminders to offer symptom checkers, telemedicine consultations, maternal health monitoring, and electronic record keeping. A landmark review by Batrachian et al. (2014) analyzed 25 randomized controlled trials of mHealth interventions in LMICs and found statistically significant improvements in health behavior outcomes when interventions were designed for the specific communication channels and literacy levels of target populations.

However, the same review highlighted a persistent problem: many mHealth apps are designed in high- resource settings by developers who have never spent a day conducting a field visit in rural Bihar or walking unpaved paths between villages in Tamil Nadu. These tools often assume stable internet connections, comfortable typing speeds on smartphones, and the ability to read English or at least a Romanized local language script. When deployed in real field conditions, they frequently fail — not because the health workers reject technology, but because the technology does not respect the conditions in which those workers operate. Care Connect was built with this failure mode explicitly in mind. Each design choice – offline data storage, voice-based input, simple forms, and multiple languages – was made based on one question: What would happen if someone used this tool during a house call to a patient in a remote village where there's no mobile network, and it's 45 degrees outside?

3.2 Offline-First Architecture in Healthcare Applications

Offline-first is a term coined by Nolan Lawson in 2013, stating that the internet should be used as an add-on, not as an integral part of the web application process (Lawson, 2013). An offline-first application will work even without an internet connection, as all operations will be performed locally on the device, with data synchronization taking place opportunistically when connected to the network. Most consumer products are equipped with an offline-first feature that enables you to browse through your news feed on social media even if you are on the subway without any signal at all. In community health technology, it is a clinical safety feature. If an ASHA worker cannot save a patient record because her mobile data has run out, that information may never be recorded. If a PHC staff member cannot access a patient's history because the local cellular tower is down, they may make a clinical decision without complete information. A study by Fraser and colleagues (2019) examined the deployment of an offline-capable community health app in rural Ethiopia and found that 34% of data entry sessions occurred in conditions of zero connectivity. In those sessions, the offline functionality of the application was not a backup — it was the primary mode of use. Similar findings have been reported from rural India, where a survey of ASHA workers in Madhya Pradesh found that 61% reported experiencing connectivity failures during at least half of their weekly home visits (Sharma et al., 2020). Care Connect implements offline functionality through the browser's local Storage API, which allows structured data to be persisted on the device without any server involvement. When connectivity is detected — using the browser's native navigator. Online event and fetch-based reachability checks — the application automatically processes the offline queue and synchronizes pending records. A visible sync counter gives the health worker transparency over what data is pending, which is important for building trust in the system. Workers who do not understand where their data is going are understandably reluctant to rely on digital tools.

3.3 Voice Interface Design for Low-Literacy and Field Users

One of the most compelling findings in human- computer interaction research for low-resource settings is that voice — not text — is the most natural and accessible input modality for a large proportion of potential users. Medhi and colleagues at Microsoft Research India (2011) conducted a series of

usability studies with illiterate and semi-literate users in India and found that voice interfaces dramatically reduced task completion times and error rates compared to text-based alternatives. Even users who had never interacted with a digital device before were able to navigate voice-driven menus and dictate information with minimal instruction. The implications for community health tools are profound: voice-enabled data entry forms have essentially eliminated the keyboard as a bottleneck. In India, the multilingual situation is even more complicated. There are 22 official languages in India, over 100 regional dialects, and a culture of multilingualism where one family can speak different languages within the home, workplace, and social settings. When designing a community health application in English, which is what all applications do, you have automatically disqualified many potential ASHA worker users from communicating in their native language. Even designing in Hindi leaves out many people in Tamil Nadu, Andhra Pradesh, Karnataka, West Bengal, and Maharashtra. The research conducted by Acharya et al. (2019) on the effectiveness of voice-based health education content among rural women of Odisha revealed that comprehension levels were 68% greater if content was presented in mother tongue rather than Hindi, and even 91% greater than English. From a practical point of view, this means that language should not be treated as some cosmetic factor of application development for health purposes but as its core element defining if the information will be understood, accepted, and put into practice. Voice input options for each mandatory text field (patient name, his/her village, symptoms) and support of seven most common Indian languages by means of Web Speech API are included in Care Connect. Web Speech API, working with no additional installation or internet connection needed in native Chrome on Android, has been used for creating a global option of voice input in which an ASHA worker can verbally tell all about the patient and the system parses the information into necessary fields based on keywords.

3.4 Electronic Health Records in Primary and Community Care

Evidence supporting the use of EHRs in primary care is quite plentiful and overwhelmingly supportive. A systematic review carried out by Chaudhry et al. (2006) showed that health IT, including EHRs, had a favorable effect on the qualit of care delivery, specifically through clinical decision support, reminders, and data structuring. Later research from low- and middle-income nations has supported these findings while stressing the importance of proper implementation in addition to the technology itself (Hersh et al., 2015). What allows EHR systems to work in primary care practices is not the complexity of software but rather how well the software matches the workflow of the user. In community-based health programs, this match tends to be poor. User interfaces developed with hospitals or clinics in mind contain too many fields, too much medical terminology, and no help in understanding what one needs to do with all of the information being gathered. According to a study carried out by Agarwal et al., 2016, health workers used the system more often when it provided immediate feedback such as calculating a risk score or suggesting a schedule for the next ANC visit. Simple act of recording some information became meaningful when the system reacted to that input with something that could be done about it. That is where the idea behind the risk classification and reminders in Care Connect came from. The issue of structured vs. unstructured information is also important here. All community health data is unstructured; it includes symptom description, results of home visits, comments about patient's environment. Molding such a narrative story into the confines of a structured field format may prove to be awkward, leading to loss of data due to workers' inability to fill up the required fields. The Care Connect tool uses an innovative solution called the voice-to-field mapping technique that enables workers to express themselves naturally and get their spoken content mapped onto the correct fields.

4. System Architecture

4.1 Technology Stack

Care Connect is built as a full-stack JavaScript application within a monorepo structure managed by pnpm workspaces.

Layer	**Technology**
Frontend	React 18 + Vite (mobile-first, PWA-ready)
Backend	Node.js with Express 5
Database	PostgreSQL with Drizzle ORM
API Contract	OpenAPI 3.1 with Orval
Voice Input	Web Speech API (SpeechRecognition)
Forms	React Hook Form + Zod Validation
Animations	Framer Motion
Styling	Tailwind CSS with Custom Healthcare Theme
Charts	Recharts

Layer	Technology
AI Assistant	OpenAI GPT Integration

4.2 Offline-First Design

The Care Connect offline solution has the following four layers:

1. Offline Queue: If the user tries to submit a record without access to the internet, the record will be stored in the browser's local storage using a specific key named 'offline patients'.

2. Connectivity Listener: The app listens to the online/offline event of the browser's navigator. When offline mode is activated, the application displays a banner informing the user about it.

3. Automatic Record Synchronization: On gaining connectivity, the app automatically reads all the pending offline records and submits each record sequentially until all the records are synchronized.

4. Synchronization Status: The app provides a real- time status indicator displaying the number of pending records and animating while synchronizing.

5. Features

5.1 Patient Data Management

A health worker may register a new patient through the following entries:

- Full name, age, and village
- Haemoglobin concentration (g/dL)
- Blood pressure (S/D)
- Pregnant or not?
- Symptoms

The system automatically calculates risk levels using rule-based logic.

Risk Classification

Condition	Risk Level
Haemoglobin < 8 g/dL	High
Haemoglobin 8–10 g/dL	Medium
Pregnant + Haemoglobin < 9 g/dL	High
BP > 140/90	High
BP 130–140 / 85–90	Medium

The risk levels are coded using colored badges in the patient cards and drive the dashboard summary statistics.

5.2 Voice Input in Regional Languages

Microphone icon appears against each text field within the patient information sheet. Once clicked, the application invokes the browser’s Speech Recognition API and proceeds to record the spoken words of the health care professional into that particular field.

Supported languages include:

- English (en-IN)
- Hindi (hi-IN) — हिंदी

- Marathi (mr-IN) — मराठी
- Tamil (ta-IN) — தமிழ்
- Telugu (te-IN) — తెలుగు
- Kannada (kn-IN) — ಕನ್ನಡ

The top section of the form provides a language selection menu, allowing the user to select their language prior to speaking. Once voice input starts, the microphone icon begins its pulse animation. The app also offers a global voice input feature, whereby the health worker can speak an entire patient profile (for example, “Patient name Sunita, age 28, village Rampur, hemoglobin8.5, pregnant yes”).

5.3 AI Health Assistant

There is a microphone button on each field of the patient form. Upon clicking this button, the application uses the browser's Speech Recognition API to transcribe the speech of the health worker into the respective field. A language selection option at the top of the patient form lets the health worker select the language they wish to use. While in voice-recording mode, there is an animated pulsating effect to indicate that the system is recording the voice of the worker. Additionally, the system has a global voice entry mode where the worker can give the description of a patient's profile in natural speech, such as, "Patient name Sunita, age 28, village Rampur, hemoglobin8.5, pregnant yes," and the system will automatically fill out the relevant fields. An assistant powered by OpenAI's GPT-5.2 model is built into the application. The health worker can type in or speak questions in their local language and get helpful health advice in response. Examples of questions include "What should I do if haemoglobin is below 8?" "What are the signs of a high-risk pregnancy?" "How do I detect malnutrition in children under 5?" The assistant is programmed using a system prompt to be culturally aware.

5.4 Vaccination and ANC Reminders

The reminders algorithm generates customized health schedules for every individual, taking into account their age and whether they are pregnant, utilizing the UIP schedule of India. For children aged less than five years, the system monitors the following vaccine schedules:

- Birth: BCG, OPV0, Hepatitis B
- 6 weeks: DPT1, OPV1, Hepatitis B2, Hib1, Rotavirus1, IPV1
- 10 weeks: DPT2, OPV2, Hib2, Rotavirus2
- 14 weeks: DPT3, OPV3, Hib3, Rotavirus3, IPV2 9 months: Measles/MR + Vitamin A
- 15 months: MMR, OPV booster, DPT booster, Vitamin A2
- 2 years: DPT2 booster, OPV booster 5 years: DT + TT

For expectant mothers, the system schedules four ANC visits in accordance with standard trimester milestones and provides alerts in the week preceding each scheduled visit.

All alerts are grouped under three alert levels:

- Immediate – Overdue or to be done today
- To Be Done This Week – In the coming 7 days
- Pending – To be done in the coming month
- High-risk patients also get automated visit alerts in seven days of their last entry date.

5.5 Multilingual User Interface

Besides voice inputs, the whole interface of the app can be changed into any of the five languages including English, Hindi, Tamil, Telugu, and Marathi.

All the navigation bars, dashboard labels, action buttons, and notifications can be instantly translated into their native language.

5.6 Data Export

Export of patient data in CSV format is possible by simply clicking once on the Patient List screen, which facilitates analysis, printing, and even distribution to your superiors or district health authorities that do not have access to the software.

6. Implementation and Results

The Care Connect full-stack implementation used a PostgreSQL database and Node.js/Express API to build a complete solution. Frontend development was done as a static website application, deployed for mobile browser accessibility without having to be installed from any app store. The system was seeded with representative patient records data containing ten patients. Risk detection, reminder generation, and visualization were working well to demonstrate the features' functionality. The reminder engine was able to identify ANC visits of pregnant women and children according to their registered ages for vaccination.

The voice input capability, developed using the Web Speech API, was able to transcribe Hindi and Marathi voice input from an Android device's Chrome browser. It will depend on the field environment and speaker characteristics whether voice input will work successfully in the real-world scenario. Offline synchronization was tested by disconnecting from the network, submitting a patient record through the form, reestablishing connection, and verifying automatic upload of the record to the database.

7. Discussion

7.1 Designing for Field Conditions

Care Connect had some of its key design decisions made due to contextual reasons rather than technical ones. For example, it made sense to allow voice input because typing on a mobile phone in an environment where an ASHA worker is visiting homes is both time consuming and inconvenient. It also made sense to automatically show risk levels because ASHA workers are trained to take actions, but they may struggle with understanding when it is necessary. Offline-first approach became relevant due to the acknowledgment that connectivity should not be taken for granted. These choices imply an underlying principle that technology meant for healthcare in rural areas should be tailored to the realities of field work and not urban development environments.

7.2Future Work

Several improvements have been envisaged in subsequent versions:

- ABHA Integration: Linking patient data with the Indian ABHA ID for creating health records across India.
- Biometric Login: Using fingerprint identification for logging into the app at the field level.
- SMS Alerts: Sending SMS to inform the patient about vaccinations and ANC in case of lack of frequent accessibility of the ASHA worker.
- Supervisor's Dashboard: This higher-end version of analytics dashboard for monitoring various health metrics of villages through various ASHA workers by district-level health officers.
- Progressive Web App (PWA): The implementation of PWA using service workers and web application manifest.

8. Conclusion

Rural and remote locations face numerous barriers in delivering health care services, including poor infrastructure, inadequate use of digital tools, and poor internet connectivity. ASHA staff are important for providing primary health services to rural communities and helping them access the healthcare system. But the conventional practice of using paper records to document the information of patients leads to various problems such as loss of information, lengthy documentation processes, and inconvenience in searching the information. Mobile- based EHR companion system is a digital solution to overcome the challenges and collect health data even from places where there is low network connectivity. ASHA personnel can enter patient data into the mobile application and store it locally if no internet connection is available, and when the device connects to the internet again, the data will be automatically synchronized.

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